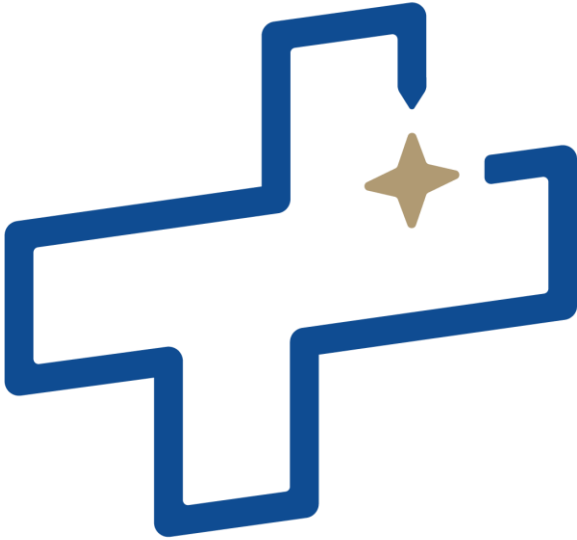




CUBESPACE

Product Description CubeTorquer

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Revision History

VERSION	AUTHORS	DATE	DESCRIPTION
1.00	C. Leibbrandt	28/07/2023	First release
2.00	C. Mailer	24/11/2025	Addition of SmallSat CubeTorquer range, updated component specifications based on manufacture data, general improvements.

Reference Documents

The following documents are referenced in this document.

[RD1]	CS-DEV.UM.CR-01	CubeTorquer User Manual Ver.2.00 or later
[RD2]	CS-DEV.ICD.CR-01	CubeTorquer NanoSat Range ICD Ver.1.03 or later
[RD3]	CS-DEV.ICD.CR-02	CubeTorquer SmallSat Range ICD Ver.1.00 or later



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1. Introduction

This document presents and describes the standalone CubeTorquer product range shown in Figure 1 which may be integrated with a client satellite system. This document is a prelude to the CubeTorquer ICD (see [RD2] and [RD3]) and standard CubeTorquer user manual (see [RD1]) which provide further detail. This CubeTorquer product description serves as an introduction to the product and documents its features, characteristics and capabilities.

Different client scenarios are catered for, namely:

1. Purchasing of a standard CubeTorquer offering by a knowledgeable client who requires no further assistance.
2. Purchasing of a CubeTorquer where the client initially requires CubeSpace consultation and suggestions to be able make an informed decision on which CubeTorquer variant to choose to optimally fulfil the client's requirements.

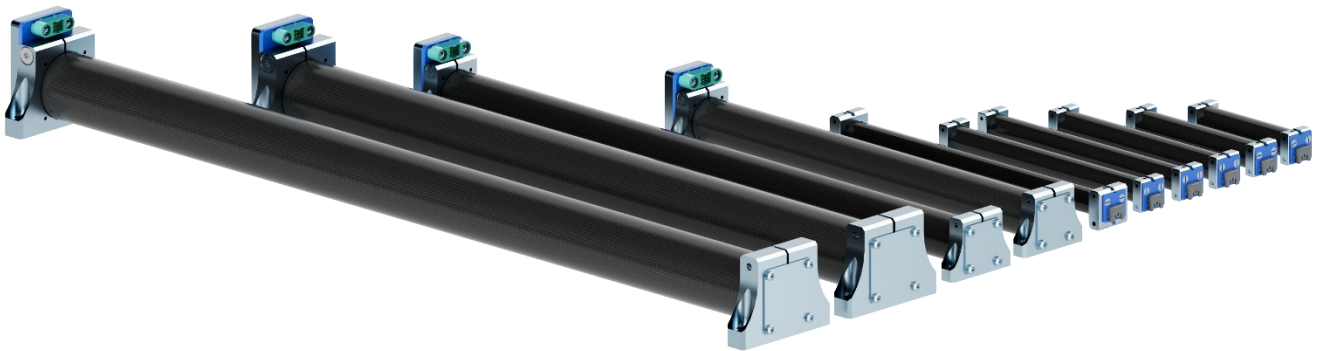


Figure 1: CubeSpace CubeTorquer Product Range

1.1 Document applicability

This version of the document applies to the CubeTorquer hardware versions as stated in Table 1 below.

Table 1: Document applicability

Variant	Version	Notes
CR0002	M2.0E1.0	M2.0E1.0: Alodine conversion coating (gold colour) M2.1E1.0: SurTec 650 conversion coating (silver colour)
CR0004	M2.1E1.0	
CR0006		
CR0010		
CR0012		
CR0020		
CR0050	M2.0E1.0	None.
CR0100		
CR0150		
CR0200		



2. CubeTorquer Context

An integrated CubeADCS is made up of several subsystems, also referred to as CubeProducts.

CubeADCS, and therefore also the CubeTorquer as a subsystem of the CubeADCS, is designed with modularity in mind. CubeProducts are typically mass manufactured, resulting in short production times and increased reliability through repeatability.

The integrated CubeADCS consists of an ADCS computer (the CubeComputer subsystem) and various other subsystem sensors and actuators, also referred to as nodes, connected via harnesses. The CubeTorquer is defined as a subsystem of type Actuator.

In such an integrated CubeADCS, the satellite OBC will interface with the CubeComputer, which will, in turn, interface with the CubeTorquer described in this document. However, each CubeProduct is also offered as a standalone product and allows for direct interfacing to a client system / client ADCS / OBC, utilising the electrical, electronic, and mechanical interfacing normally utilized for interfacing to the CubeADCS.

All interfacing details for the CubeTorquer are presented in the relevant CubeTorquer Interface Control Document (see [RD2] and [RD3]).

The CubeTorquer User Manual (see [RD1]), is also provided, typically post order placement, to guide the client through initial usage, handling, and to be able to conduct a health check test after receipt of the physical item.



3. Detailed Description

3.1 Overview

CubeTorquers are magnetic torquer rods, intended as actuators used for controlling the orientation of satellites in Low Earth orbit. This component generates a magnetic field when current is passed through it, which results in a torque on the satellite through interaction with the earth's magnetic field. Typically, three CubeTorquers are used per satellite, placed in the three principal axes of the satellite.

The CubeTorquer product has been refined over a decade of use in orbit and offers excellent performance, is robust, integration friendly, and comes in a range of sizes for satellites ranging in sizes from 2U to 400kg. CubeTorquers are manufactured in a production facility that, through innovation and automation, is geared for mass manufacturing, and can be repeatably delivered in high quantities, on short lead-times, at affordable prices.



Handling Warning: CubeTorquer must be handled with care. Avoid exposing to hard impacts or high-frequency mechanical shocks, as these can alter the magnetic properties of the cores, affecting both linearity and remanence. Examples include dropping the device onto hard surfaces or striking it against objects. While CubeTorquer has been qualified through shock and vibration testing representative of satellite campaigns and launch environments, careful handling is essential to preserve optimal performance.

The main features of CubeTorquer include:

- Heat treated, high permeability, low remanence core
- High magnetic gain
- Robust mounting
- Low profile Molex Pico-Lock connector, or high reliability Harwin Gecko SL
- Large variant range suitable for Satellite sizes from 1U up to 400kg.
- Kapton and Carbon Fibre protective covering for windings

CubeTorquers are built using automated machinery and go through rigorous testing, which ensures absolute repeatability, and enables high volumes and low cost. An overview of the CubeTorquer is presented in Table 2.

Table 2: CubeTorquer description

Item	Description	
	Description	Magnetic Torque Rod
	Details	• Ultra-low remanence magnetic actuators • Available in various sizes from 0.2 Am² [CR0002] to 20.0 Am² [CR0200]
	Generic Term	Magnetorquer (MTQ)
	CS Name and acronym	CubeTorquer (CR)



3.2 Functions and Features

The primary function of a CubeTorquer is to generate a magnetic dipole when, and only when, current is passed through it. Electrically conductive wire is wound around a permeable core. The core is made from heat-treated ferrous material, which has a high permeability, high saturation magnetic flux, and ensures a low remnant magnetic field.



CubeTorquer is typically driven from a voltage source, although some systems use variable-current drivers. The magnetic moment is controlled by adjusting the average current, either by varying the duty cycle (average on-time) or by changing the supply voltage. The design of CubeTorquer ensures a highly linear relationship between supply current and magnetic moment, with very **low total linearity error**, allowing control algorithms to assume linear behaviour.

A key performance factor of CubeTorquer is their **remnant magnetic moment** - the residual magnetization that remains in the core material after the coil has been energized and switched off. If this remnant field is too high, it can interfere with onboard magnetometers, generate unwanted disturbance torques, and affect sensitive satellite payloads. CubeTorquer is designed to exhibit a negligible remnant magnetic moment, ensuring minimal interference and reliable satellite performance.

CubeTorquer is offered with **short lead-times** and **global availability** making it very simple to integrate into any satellite mission.

3.3 Performance characteristics

Table 3: CubeTorquer specifications and performance characteristics

CUBETORQUER VARIANTS										
	CRO002	CRO004	CRO006	CRO010	CRO012	CRO020	CRO050	CRO100	CRO150	CRO200
PERFORMANCE										
Dipole @ 5V [Am²]	0.2	0.4	0.6	1.0	1.2	2.0	5.0	10.0	15.0	20.0
Magnetic Gain [Am²/A]	2.3	3.3	5.6	7.6	10.1	13.9	22.0	46.0	63.0	74.5
Linearity Error	<2.0% of FSR									
Residual Dipole	<0.1% of FSR									
Resistance [Ω]	51.0	39.5	45.0	37.5	36.5	32.5	20.9	21.8	19.9	17.7
Inductance [mH]	120	180	270	310	233	260	490	750	790	650
Time Constant [ms]	2	5	6	8	6	8	23	34	40	37
PHYSICAL										
Mass [g]	16	23	31	37	45	54	164	272	598	743
Length [mm]	47	59	77	92	122	152	167	247	277	337
POWER & DATA										
Rated Voltage [V]	5.0									
Connector	Molex Pico-Lock						Harwin Gecko SL			
QUALIFICATION LEVELS										
Vibration	14.16 G RMS (NASA GEVS)									
Shock	1500 G									
Thermal vacuum	-20°C to 80°C									
Thermal cold & hot start	-35°C to 70°C									



3.4 CubeTorquer actuator sizing

The selection of CubeTorquer variants must be determined based on the satellite's size, deployable structures, and required manoeuvrability.

CubeTorquer is available in a range of sizes and can be selected based on mission needs. For 6U and smaller satellites, up to 3 magnetorquers should be considered. The number of magnetorquers for use as standalone subsystems within a client satellite is however unlimited. Table 4 provides a rough guideline for selecting between the various actuator sizes available for CubeTorquer depending on satellite size and desired performance.

Table 4: Actuator specifications

Satellite size	CubeTorquer	Mounting
Up to 6U	0.2 Am ² to 0.4 Am ² [CR0002 to CR0004]	CubeTorquers can be mounted externally or internally.
6U to 27U	0.4 Am ² to 2.0 Am ² [CR0004 to CR0020]	
50kg to 100kg	5.0 Am ² [CR0050]	
100kg to 200kg	10.0 Am ² [CR0100]	
200kg to 300kg	15.0 Am ² [CR0150]	
300kg to 400kg	20.0 Am ² [CR0200]	

3.5 Interconnect

The CubeTorquer actuator is designed to be connected to the CubeADCS CubeComputer or to the client system by means of wire harnesses. These harnesses are made using low-outgassing insulation. All CubeTorquer interface related information is detailed in the relevant CubeTorquer ICD (see [RD2] and [RD3]).

3.6 CubeTorquer coordinate systems.

The CubeTorquer actuators implement their own Local Coordinate Frame defined in the relevant CubeTorquer Interface Control Document (see [RD2] and [RD3]).

3.7 CubeTorquer actuator placement

In general, all CubeSpace's actuators, and specifically the CubeTorquer must be placed in locations far enough from sensitive payloads or sensors that can be disturbed by their magnetic fields.

CubeTorquers exhibit very small residual dipoles, but care should still be taken to mount them as far as possible from magnetically sensitive payloads, magnetometer sensors, and magnetic materials. Although magnetometer measurements should only be taken during windows of CubeTorquer inactivity, they can still be disturbed by the residual magnetic moments and the magnetic field decay time.

3.8 Harness lengths

Harness lengths are typically not known until detailed satellite layout has been decided on. CubeSpace will supply a detailed wiring / harness list documenting all aspects of harnesses to the CubeTorquer. CubeSpace will request the client to indicate harness lengths based on the final placement of the CubeTorquer subsystems within the client satellite/system and the clients harness routing scheme.



3.9 Documentation

The CubeTorquer is provided with a set of standard documents which are listed in Table 5:

Table 5: CubeTorquer standard documentation

Document	Description
Standalone Product Description (PD) - (This Document)	Provides an overview of the standard CubeSpace CubeTorquer offering. It is typically supplied to prospective clients to allow a better understanding of the CubeSpace CubeTorquer offering.
Standalone Interface Control Document (ICD)	Detailed information about the physical aspects of the standard CubeTorquer offering addressing technical aspects of all interfaces. It is typically supplied to prospective clients to allow a better understanding of the CubeSpace CubeTorquer offering and how to interface to it; electrically, mechanically and electronically.
Standalone User Manual (UM)	Describes all functions and features in more detail (than provided in the Product Description). It also allows the user to conduct a Health Check to confirm the CubeTorquer is “alive and well” after receipt of the shipment at the client. It is typically only supplied to actual clients.
Delivery Report (DR)	Report to indicate that QA took place on delivered unit and that a complete health check was performed at CubeSpace before packaging and shipment.